

Swissair Flight 111

Flight model: Douglas MD-11
Date: September 02, 1998

From: New york, US
To: Geneva, Switzerland

Problem

The flight took off normal. The first 15 minutes after take off there were not communication in the plane which is unusual. The planed path was a trans-atlantic crossing. The plane had for the very first time introduced in flight entertainment in its fleet. This plane was one among the few which had an in-flight entertainment. Soon the pilots smell smoke in the cockpit. They assume that the possible reason would be due to air conditioning and turn off the same. After some time, the pilots notices that the smoke increase and decides to deviate the flight to a nearest airport to land. A "PAN-PAN-PAN" was declared to the nearest ATC. The crew had identified the need to dump fuel to be able to land successfully. The ATC advice the crew to perform a circle approach to dump the fuel above the sea while staying close to the airport. The pilots then turn the **cabin-bus** switch off, this turn off the cabin light; this is done to indicate the passengers regarding the situation while not panicking them at the same time. Suddenly alarm indicating a serious problem was raised at the same time the was visible fire in the cabin. Now, an emergency was declared. All screens of the pilots were down. Later, the ATC tried to communicate with the plane but was not able to do so. Later, the plane crashed into the ocean.

Investigation

While analyzing the CVR, it was found that the last 6 minutes of the data was corrupt and was in a format that cannot be read. The scoot marks on the wiring material revels the possibility of fire due to electrical arching. The suspect here was the in-flight entertainment which drove a large amount of power. Also, there were no kill switch for the in-flight entertainment were the pilots could turn them off when needed. It was found that the wire ignited the insulation material and the insulation material was responsible for the spread of the fire.

Changes that were brought after the incident

The report advises the manufactures the remove the flammable materials from the wiring panel.

Reference

1. Swissair 111's Last Minutes — Mayday Air Disaster
<https://www.youtube.com/watch?v=Z42Cu6oq9qQ>